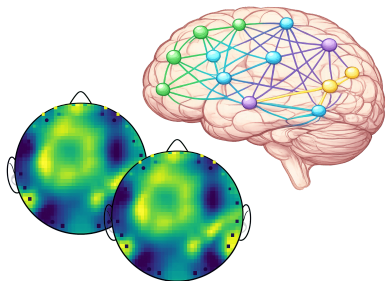


Contribution I

EEG-GCIRNet

Gaussian Functional Connectivity

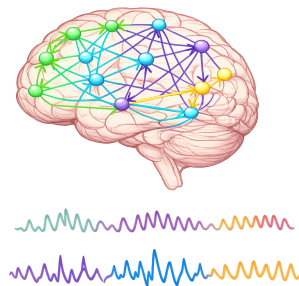


- Spatial-Structural Preservation
- EEG Imaging
- Connectivity-Driven Representation
- Noise & Volume Conduction Robustness

Contribution II

TEKTE-Net

Directed & Nonlinear Effective Connectivity



- Takens State-Space
- Kernel Transfer Entropy
- Directed Connectivity
- Nonlinear Dynamics

Contribution III

CTE-Net

Transformer-Contextualized Directed Connectivity



- Channel-Wise Temporal Filtering
- Takens Delay-coordinate Reconstruction
- Takens-Based Differentiable TE

Deep Learning for Robust EEG Decoding in Brain-Computer Interfaces